

Windows Forensics 1

TryHackMe Lab Documentation

Introduction to Windows Registry Forensics

Room: <https://tryhackme.com/room/windowsforensics1>

Task 1: Introduction to Windows Forensics

Overview

Computer forensics is the practice of collecting and analyzing digital evidence from computers to investigate crimes or incidents. Windows holds approximately 80% of the global desktop market share, making it the most commonly encountered operating system in forensic investigations.

A forensic artifact is a small trace or footprint left behind on a computer after a user performs an action. These artifacts are used by investigators to reconstruct a timeline of events. A real-world example is the BTK killer case, where investigators recovered a deleted Word document from a floppy disk. The metadata in that document helped identify and arrest the suspect.

What is the most used Desktop Operating System right now?

Answer: Microsoft Windows

Task 2: Windows Registry and Forensics

Overview

The Windows Registry is a hierarchical database that stores configuration data for the operating system, hardware, software, and user preferences. It is made up of Keys (folders), Values (data inside folders), and Hives (groups of keys stored as a single file on disk).

The Five Root Keys

Root Key	Short Form	Description
HKEY_CURRENT_USER	HKCU	Settings for the currently logged-in user
HKEY_USERS	HKU	Settings for all user profiles on the computer
HKEY_LOCAL_MACHINE	HKLM	Hardware and software config for the whole computer
HKEY_CLASSES_ROOT	HKCR	File association and COM object registration

HKEY_CURRENT_CONFIG	-	Hardware profile used at system startup
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What is the short form for HKEY_LOCAL_MACHINE?

Answer: HKLM

Task 3: Accessing Registry Hives Offline

Overview

When analyzing a disk image rather than a live system, you must know the physical location of registry hive files on disk. The five main hives are stored in C:\Windows\System32\Config. User-specific hives are located in the user profile directory.

Registry Hive Locations

Hive File	Location	Mounted To
DEFAULT	C:\Windows\System32\Config	HKEY_USERS\DEFAULT
SAM	C:\Windows\System32\Config	HKEY_LOCAL_MACHINE\SAM
SECURITY	C:\Windows\System32\Config	HKEY_LOCAL_MACHINE\Security
SOFTWARE	C:\Windows\System32\Config	HKEY_LOCAL_MACHINE\Software
SYSTEM	C:\Windows\System32\Config	HKEY_LOCAL_MACHINE\System
NTUSER.DAT	C:\Users\<username>\	HKEY_CURRENT_USER
USRCLASS.DAT	C:\Users\<username>\AppData\Local\Microsoft\Windows	HKEY_CURRENT_USER\Software\Classes
Amcache.hve	C:\Windows\AppCompat\Programs\	Tracks recently run programs

Transaction logs (.LOG1, .LOG2) store the latest registry changes not yet written to the hive. Registry backups are stored in C:\Windows\System32\Config\RegBack and are updated every 10 days.

What is the path for the five main registry hives?

Answer: C:\Windows\System32\Config

What is the path for the AmCache hive?

Answer: C:\Windows\AppCompat\Programs\Amcache.hve

Task 4: Data Acquisition

Overview

Data acquisition is the process of making a forensic copy of data before performing analysis. This ensures the original evidence is not altered. Registry hive files in C:\Windows\System32\Config are locked by the OS and cannot be directly copied, so specialized tools are required.

Acquisition Tools

Tool	Use Case	Notes
KAPE	Live data acquisition	Command-line and GUI. Fast and widely used
Autopsy	Live system or disk image	Right-click Extract File(s) option
FTK Imager	Live system or disk image	Obtain Protected Files option (does not copy Amcache.hve)

Note: The FTK Imager Obtain Protected Files option extracts all registry hives except Amcache.hve, which must be acquired separately.

Answer: Practical exercise. No answer required.

Task 5: Exploring Windows Registry

Overview

Once registry hives are extracted, specialized tools are needed to view and analyze them since regedit.exe only works on live systems.

Analysis Tools Comparison

Tool	Strengths	Limitations
Registry Viewer (AccessData)	Similar interface to regedit	One hive at a time, ignores transaction logs
Registry Explorer (Zimmerman)	Multiple hives, merges transaction logs, has bookmarks	None significant
RegRipper	Auto-extracts forensic data, outputs text report	Ignores transaction logs

Best Practice: Use Registry Explorer to merge transaction logs into hives first, then pass the clean hive to RegRipper for the most accurate results.

Answer: Study exercise. No answer required.

Task 6: System Information and System Accounts

Overview

The first step in forensic analysis is establishing basic system facts. This includes the OS version, computer name, time zone, network configuration, and user accounts. All of this information is found within the registry hives.

Key Registry Paths

Information	Registry Path
OS Build Number	SOFTWARE\Microsoft\Windows NT\CurrentVersion
Current Control Set	SYSTEM\Select\Current
Last Known Good Config	SYSTEM\Select\LastKnownGood
Computer Name	SYSTEM\CurrentControlSet\Control\ComputerName\ComputerName
Time Zone	SYSTEM\CurrentControlSet\Control\TimeZoneInformation
Network Interfaces	SYSTEM\CurrentControlSet\Services\Tcpip\Parameters\Interfaces
User Accounts	SAM\Domains\Account\Users

What is the Current Build Number of the machine?

Answer: 19044

Which ControlSet contains the last known good configuration?

Answer: ControlSet001

What is the Computer Name of the computer?

Answer: THM-4N6

What is the value of the TimeZoneKeyName?

Answer: Pakistan Standard Time

What is the DHCP IP address?

Answer: 192.168.100.58

What is the RID of the Guest User account?

Answer: 501

Task 7: Usage or Knowledge of Files/Folders

Overview

Windows silently records every file a user opens, every folder they visit, every path they type, and every location used in Open/Save dialogs. This data is invaluable for reconstructing user activity during a forensic investigation.

Key Registry Paths

Artifact	Registry Path	What It Shows
Recent Files	NTUSER.DAT\...\Explorer\RecentDocs	Recently opened files with timestamps
Office Recent Files	NTUSER.DAT\...\Microsoft\Office\VERSION	Recently opened Office documents
ShellBags	USRCLASS.DAT\...\Shell\Bags	Folder layouts and recently visited folders
Open/Save Dialogs	NTUSER.DAT\...\ComDlg32\OpenSavePIDMRU	Files opened or saved via dialog boxes
Typed Paths	NTUSER.DAT\...\Explorer\TypedPaths	Paths typed in the Explorer address bar
Search History	NTUSER.DAT\...\Explorer\WordWheelQuery	Terms searched in Windows Explorer

When was EZtools opened?

Answer: 2021-12-01 13:00:34

At what time was My Computer last interacted with?

Answer: 2021-12-01 13:06:47

What is the Absolute Path of the file opened using notepad.exe?

Answer: C:\Program Files\Amazon\Ec2ConfigService\Settings

When was this file opened?

Answer: 2021-11-30 10:56:19

Task 8: Evidence of Execution

Overview

Even after a program is deleted, Windows may retain records that it was executed. There are four key artifacts for tracking program execution. These are critical for proving that a suspect ran a particular program.

Execution Artifacts

Artifact	Registry/File Path	What It Records
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UserAssist	NTUSER.DAT\...\UserAssist\{GUID}\Co unt	Programs launched via Explorer, run count, last executed time
ShimCache (AppCompatCache)	SYSTEM\CurrentControlSet\Control\Session Manager\AppCompatCache	All applications launched, file name, size, last modified time
AmCache	C:\Windows\AppCompat\Programs\Am cache.hve	Execution path, install/delete times, SHA1 hashes
BAM/DAM	SYSTEM\CurrentControlSet\Services\lsass\UserSettings\{SID}	Full path of executed programs and last execution time

How many times was File Explorer launched?

Answer: 26

What is another name for ShimCache?

Answer: AppCompatCache

Which artifact saves SHA1 hashes of executed programs?

Answer: Amcache

Which artifact saves the full path of executed programs?

Answer: BAM/DAM

Task 9: External Devices / USB Device Forensics

Overview

When a USB device is connected to a Windows machine, the registry records detailed information about it. Even after the device is removed, this evidence remains. This is particularly useful for detecting data exfiltration or unauthorized device usage.

Key Registry Paths

Information	Registry Path
USB Device Identification	SYSTEM\CurrentControlSet\Enum\USBSTOR

USB Vendor/Product Info	SYSTEM\CurrentControlSet\Enum\USB
Device Volume Name	SOFTWARE\Microsoft\Windows Portable Devices\Devices

Connection Time Codes (within USBSTOR key)

Code	Information
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0064	First connection time
0066	Last connection time
0067	Last removal time

What is the serial number of the Kingston device?

Answer: 1C6F654E59A3B0C179D366AE&80

What is the name of the Kingston device?

Answer: Kingston DataTraveler 2.0 USB Device

What is the friendly name of the Kingston device?

Answer: USB

Task 10: Hands-on Challenge

Scenario

One of the Desktops in the research lab at Organization X was suspected to have been accessed by an unauthorized user. Multiple user accounts were observed on the system, and it was suspected that the system was connected to a network drive with a USB device also connected. Triage data was collected using KAPE and placed on the attached VM for analysis.

Setup

Item	Details
VM Credentials	Username: THM-4n6 Password: 123
Triage Folder	C:\Users\THM-4n6\Desktop\triage (KAPE collection)
Tools Folder	C:\Users\THM-4n6\Desktop\EZtools (Registry Explorer, EZViewer, AppCompatCacheParser)
Primary Tool Used	Registry Explorer (Eric Zimmerman)

Investigation Methodology

Registry hives were loaded from the triage folder into Registry Explorer. When prompted about dirty hives, transaction logs (.LOG1 and .LOG2) were merged to create clean hives for accurate analysis. The SAM hive was used for user account information, NTUSER.DAT for user activity, and the SYSTEM hive for USB device evidence.

Questions and Answers

Q1: How many user created accounts are present on the system?

Answer: 3

Registry Path: SAM\Domains\Account\Users\Names

Method: Loaded the SAM hive from triage\C\Windows\System32\Config\SAM into Registry Explorer. Navigated to SAM\Domains\Account\Users\Names. Identified 7 total accounts and removed the 4 built-in defaults (Administrator, DefaultAccount, Guest, WDAGUtilityAccount), leaving 3 user-created accounts: THM-4n6, thm-user1, and thm-user2.

Q2: What is the username of the account that has never been logged in?

Answer: thm-user2

Registry Path: SAM\Domains\Account\Users

Method: Used the net user command in Command Prompt on the VM. Running 'net user thm-user2' showed Last logon: Never, confirming this account had never been logged into.

Q3: What is the password hint for the user THM-4n6?

Answer: count

Registry Path: SAM\Domains\Account\Users\000003E9

Method: In Registry Explorer, clicked on RID 000003E9 (THM-4n6 account). Selected the UserPasswordHint value in the right panel. The Type Viewer at the bottom of Registry Explorer decoded the binary data to reveal the password hint: count.

Q4: When was the file 'Changelog.txt' accessed?

Answer: 2021-11-24 18:18:48

Registry Path: NTUSER.DAT\Software\Microsoft\Windows\CurrentVersion\Explorer\RecentDocs Method: Loaded NTUSER.DAT from triage\C\Users\THM-4n6 into Registry Explorer. Navigated to the RecentDocs key and located Changelog.txt in the list. The Opened On timestamp showed 2021-11-24 18:18:48.

Q5: What is the complete path from where the Python 3.8.2 installer was run?

Answer: Z:\setups\python-3.8.2.exe

Registry Path: NTUSER.DAT\Software\Microsoft\Windows\CurrentVersion\Explorer\UserAssist

Method: In Registry Explorer with NTUSER.DAT loaded, navigated to the UserAssist key. Located the Python 3.8.2 installer entry and noted the full execution path: Z:\setups\python-3.8.2.exe. The Z: drive indicates the installer was run from a network drive, which aligns with the scenario of the system being connected to a network drive.

Q6: When was the USB device with the friendly name 'USB' last connected?

Answer: 2021-11-24 18:40:06

Registry Path: SYSTEM\CurrentControlSet\Enum\USBSTOR

Method: Loaded the SYSTEM hive from triage\C\Windows\System32\Config into Registry Explorer. Navigated to the USBSTOR key and located the Kingston DataTraveler device with friendly name USB. The Last Connected timestamp showed 2021-11-24 18:40:06.

Conclusion

This lab demonstrated how Windows Registry forensics can be used to reconstruct the full picture of activity on a compromised system. By analyzing various registry hives using Registry Explorer, it was possible to:

- Identify unauthorized user accounts created on the system
- Determine which accounts had never been used
- Recover password hints stored in the SAM hive
- Track file access timestamps through RecentDocs
- Identify that a network drive (Z:) was connected and used to run a Python installer -
Confirm the connection of a USB device and its last connected timestamp

The Windows Registry serves as a comprehensive and detailed log of all system and user activity, making it an invaluable resource for digital forensic investigators. Every action leaves a trace, and with the right tools and knowledge, those traces tell the full story.